

Dr. Pranav Chakra Varthy Bommarasipeta

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SUMMARY

- Civil Engineer with 4 years of industry experience with bridge construction in large projects and steel design.
- Strong background in modeling and use of structural analysis software, design tools and Finite Element Modelling
- Proficient in using Python and MATLAB for numerical simulations and signal processing.
- Familiar with seismic design of buildings using ACI and AISC codes
- Ability to handle high-priority deliverables and coordinate with key stakeholders for timely execution.

PROFESSIONAL EXPERIENCE

University at Buffalo

Aug 2023 — Aug 2026

Graduate Research Assistant | Teaching Assistant (Advisor: Dr. M. V Sivaselvan)

Buffalo, NY

- Investigated the effects of protective device for electrical transformer bushings in mitigating rupture during earthquake .
- Developed FEM model of transformer in ABAQUS and wrote python scripts for automation and debugging.
- Developed AutoCAD and Autodesk Inventor models for fabrication to repurpose motion table for mini shake table tests.
- Developed analytical model for hysteretic spring action of protective device and bushing's rocking behavior in MATLAB.
- Conducted shake table tests of life-sized bushing and processed sensor data to fine-tune the numerical models.
- Taught civil engineering courses - Statics (Fall 2023, Fall 2025) and Structural Analysis (Summer 2026)

L&T Construction

Jul 2019 — Jun 2023

Sr. Engineer - Construction Methods

Chennai, India

- Planned bridge construction and delivered staging plans, fabrication drawings and design documents.
- Coordinated with on-site team for deliverables and immediate response to high-priority technical challenges.
- Worked in large project packages in India such as Eastern Dedicated Freight Corridor, Mumbai Ahmedabad High-Speed Rail and Chennai Metro Rail (Phase-II) worth atleast \$250 Million (as of 2020)
- Presented construction plans to government authorities to get approval for bridge construction over existing railway line.
- Perfected reuse of temporary structures across sites in EDFC project and saved 2 months of time and about 1 million USD compared to the initial plan.

Indian Institute of Technology, Gandhinagar

Jul 2018 — Jul 2019

Graduate research assistant (Advisor: Dr. Dhiman Basu)

Gujarat, India

- Studied the dynamic properties of confined masonry buildings and proposed force-based design recommendations for these structures against seismic action.
- Aimed to propose standards for dynamic properties of CM buildings as no such provisions were given in any Indian and International standard codes.
- Conducted ambient vibration tests on Nine (G + 3) hostels at IIT Gandhinagar that are built using the confined masonry method. Frequency Domain analysis was performed using MATLAB to obtain frequencies and mode shapes
- An empirical formula for base shear, frequency, and damping estimates was proposed as a codal provision.

PUBLICATIONS

- Abdelrahman Elmeliegy, Jon Bender, **Pranav Bommarasipeta**, Vidhi Solanki, Mettupalayam Sivaselvan, Aisha Mckee, Bjorn Vaagensmith, and Chandrakanth Boliseti. Novel Decoupler Solution for Transformer Bushing: Simulation and Experimental Confirmation (In Draft)
- **Chakra-Varthy, Pranav** & Basu, Dhiman. (2021). Natural period and vertical distribution of base shear in confined masonry buildings using ambient vibration test. Bulletin of Earthquake Engineering. 10.1007/s10518-021-01046-8.
- Kafie, Jackeline & Keating, Peter & **Chakra-Varthy, Pranav** & Correia, José & De Jesus, Abilio. (2018). Stress distributions and crack growth in riveted lap joints fastening thick steel plates. Engineering Failure Analysis. 91. 10.1016/j.engfailanal.2018.04.048.

EDUCATION

PhD in Civil Engineering

University at Buffalo

Buffalo, NY

Aug 2023 — Aug 2026

- Recipient of Graduate Fellowship from University at Buffalo from Fall-2023 to Spring-2026

Master of Technology in Civil Engineering (Integrated B.Tech-M.Tech Dual Degree)

Indian Institute of Technology, Gandhinagar

Gujarat, India

Aug 2018 — Jul 2019

Bachelor of Technology in Civil Engineering (Minors in Computer Science)

Indian Institute of Technology, Gandhinagar

Gujarat, India

Aug 2014 — Jul 2018

- Received Academic Excellence Scholarship for the year at IITGN (2016–2017)
- Awarded Dean's list for Academic Excellence at IITGN: 2015–2016 (Semester-I), 2014–2015 (Semester-I)
- Recipient of S.P Mehrotra Scholarship for the years at IITGN (2015–2018)

TECHNICAL SKILLS

- **Design Tools:** ABAQUS, LARSA 4D, SAP 2000, Staad Pro, MathCAD, AutoCAD, Autodesk Inventor
- **Programming Languages:** Python, MATLAB
- **Software & Tools:** Microsoft Office, Linux commands, Git & Agentic workflows